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calculate the percentage efficiency of an engine.

Solution:

Given

$T_h = 127^0$ C = 127 + 273 = 400 K

$T_C = 47^0$ C = 47 + 273 = 320 K

% efficiency η = ?

$$\eta = \left[\frac{T_h - T_C}{T_h}\right] \times 100$$

$$\eta = \left[\frac{400 - 320}{400}\right] \times 100$$

$$\eta = \left[\frac{80}{400}\right] \times 100$$

η = 20%

6. Urea on hydrolysis produces ammonia and carbon dioxide. The standard entropies of urea, H_2O, CO_2, NH_3 as 173.8, 70, 213.5 and 192.5 J mol^{-1} K^{-1} respectively. Calculate the entropy change.

Solution:

Given:

S^0 (urea) = 173.8 J mol^{-1} K^{-1}

S^0 (H_2O) = 70 J mol^{-1} K^{-1}

S^0 (CO_2) = 213.5 J mol^{-1} K^{-1}

S^0 (NH_3) = 192.5 J mol^{-1} K^{-1}

$NH_2 - CO - NH_2 + H_2O \rightarrow$

$2NH_3 + CO_2$

$\Delta S_r^0 = \Sigma (S^0)_{products} - \Sigma (S^0)_{reactants}$

$\Delta S_r^0 = [2\ S^0 (NH_3) + S^0 (CO_2)]$

$- [S^0 (urea) + S^0 (H_2O)]$

$\Delta S_r^0 = [2 \times 192.5 + 213.5]$

$- [173.8 + 70]$

$\Delta S_r^0 = [598.5] - [243.8]$

$\Delta S_r^0 = 354.7$ J mol^{-1} K^{-1}

7. Calculate the entropy change when 1 mole of ethanol is evaporated at 351 K. The molar heat of vaporisation of ethanol is 39.84 kJ mol^{-1}.

Solution:

Given:

T_b = 351 K

ΔH_{vap} = 39840 J mol^{-1}

ΔS_v = ?

$$\Delta S_v = \frac{\Delta H_{vap}}{T_b}$$

$$\Delta S_v = \frac{39840}{351}$$

$\Delta S_v = 113.5$ J K^{-1} mol^{-1}

8. For a chemical reaction the values of ΔH and ΔS at 300 K are – 10 kJ mol^{-1} and –20 J K^{-1} mol^{-1} respectively. What is the value of ΔG of the reaction? Calculate the ΔG of a reaction at 600K assuming ΔH and ΔS values are constant. Predict the nature of the reaction.

Solution:

Given:

$\Delta H = -10$ kJ $mol^{-1} = -10000$ J mol^{-1}

$\Delta S = -20$ J K^{-1} mol^{-1}

T = 300 K